Capstone controlled experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Capstone controlled experiments before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone hypothesis and baseline

Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Efficiency and resource measurement

Without opening the old lesson, explain one key idea from Efficiency and resource measurement. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Transfer learning

Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Continue with **training** (1,2) and (2,4), validation (3,6) and (4,8), and **prediction** max(0, a*x). Compare learning rates 0.1 and 0.2. Each run starts at a = 1, has no bias and takes two full-batch updates.

Keep the loss, data order and arithmetic fixed. This toy model has no randomness; repeating the exact run checks replay, not variation across seeds.

Record the comparison as an explicit revision to the one-candidate Week 33 plan before viewing new results. Validation may guide selection; sealed final data must not.

Select lower validation MSE, with a tie going to learning rate 0.1.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Capstone controlled experiments trace and record the first step you can explain without help.

Use W2: Trace both runs for Capstone controlled experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Capstone controlled experiments explanation against the supplied facts before discussing it.

Use the worked case for Capstone controlled experiments. Proposed explanation: “More runs mean more validation examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Run a new comparison independently

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Name the controls

> Name the controls. Apply the supplied Capstone controlled experiments facts and keep your reasoning.

List six settings held fixed and the single intended changed factor.

My reasoning and evidence

W2 Trace both runs

> Trace both runs. Apply the supplied Capstone controlled experiments facts and keep your reasoning.

Compute both updates for rates 0.1 and 0.2 using gradient 5*(a-2).

My reasoning and evidence

W3 Select without extra evidence

> Select without extra evidence. Apply the supplied Capstone controlled experiments facts and keep your reasoning.

Recompute validation MSE for the final weights and apply the written **selection rule**. How many distinct validation rows were used?

My reasoning and evidence

W4 Run a new comparison independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Run a new comparison independently. Apply the supplied Capstone controlled experiments facts and keep your reasoning.

For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1. Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Audit a mixed comparison

> Audit a mixed comparison. Apply the supplied Capstone controlled experiments facts and keep your reasoning.

Run A uses rate 0.1 for two updates; run B uses rate 0.2 for six updates and different **training** rows. Can the difference be attributed to rate alone? Propose a repair.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Capstone controlled experiments to Image-pattern Network Investigation, identify a limit and choose the next useful test.

For Image-pattern Network Investigation, use this week’s independent task as a rehearsal for the final **evidence** pack: For **training** (1,3),(2,6) and validation (3,9),(4,12), start a = 1.

Compare rates 0.1 and 0.2 for two updates. Gradient is 5*(a-3). Compute final weights and validation MSE, then select lower MSE; ties go to 0.1.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**
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WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

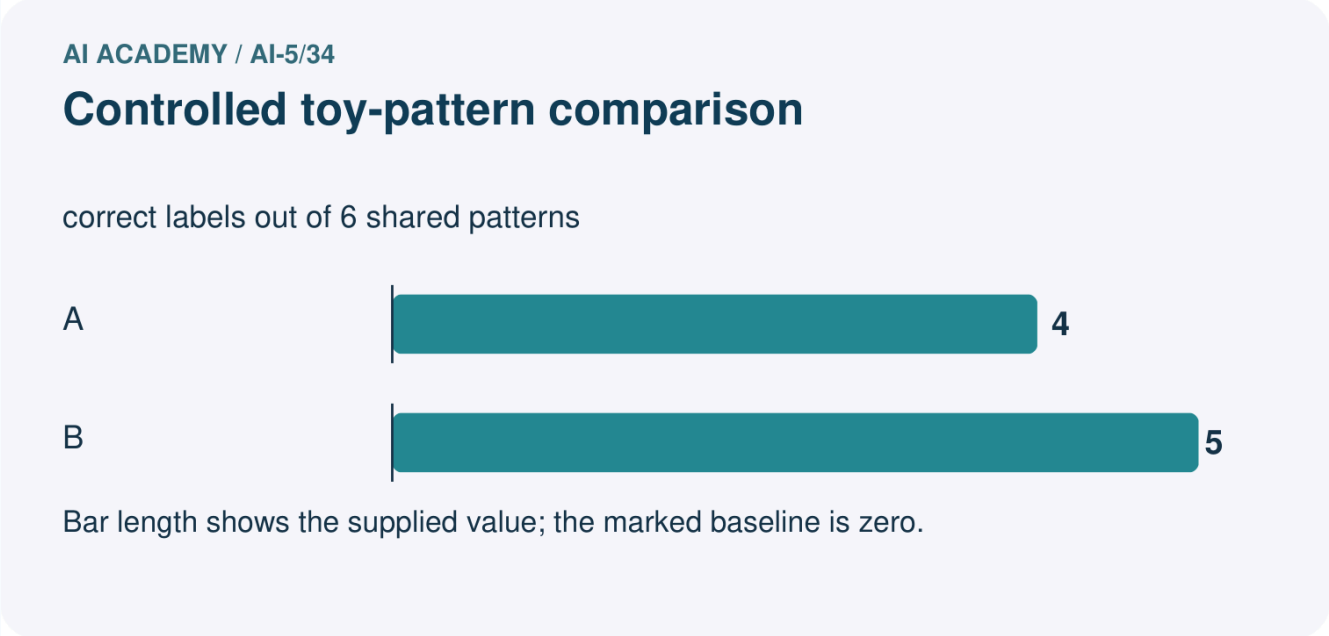
WORKBOOK · 12 OF 15

Pictorial case practice: Capstone controlled experiments

> One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added. Explain the labelled step before checking the changed condition.

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1]. Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/34

Capstone controlled experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the intervention

B adds the centre-feature connection; all other declared settings stay fixed.



2. Count outcomes

A is 4/6 correct; B is 5/6.



3. Inspect paired cases

Only case three changes correctness; case five fails for both.

CHANGE ONE THING

Change B's training set as well as the feature connection.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

State exactly what this example supports before linking it to the wider capstone claim.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Only case three changes correctness; case five fails for both.

1. Inspect paired cases

2. Identify the intervention

3. Count outcomes

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect paired cases Only case three changes correctness; case five fails for both.

Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change B's **training** set as well as the feature connection. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The experiment no longer isolates the connection change.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The experiment no longer isolates the connection change. A **training**-data difference supplies an alternative explanation for the observed result.

Repair a missing building block

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Efficiency and resource measurement (ai-5/31). Goal: Calculate throughput and amortised time per example from repeated timings. Short practice: Why is 10 ms/case not necessarily a single-request latency?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

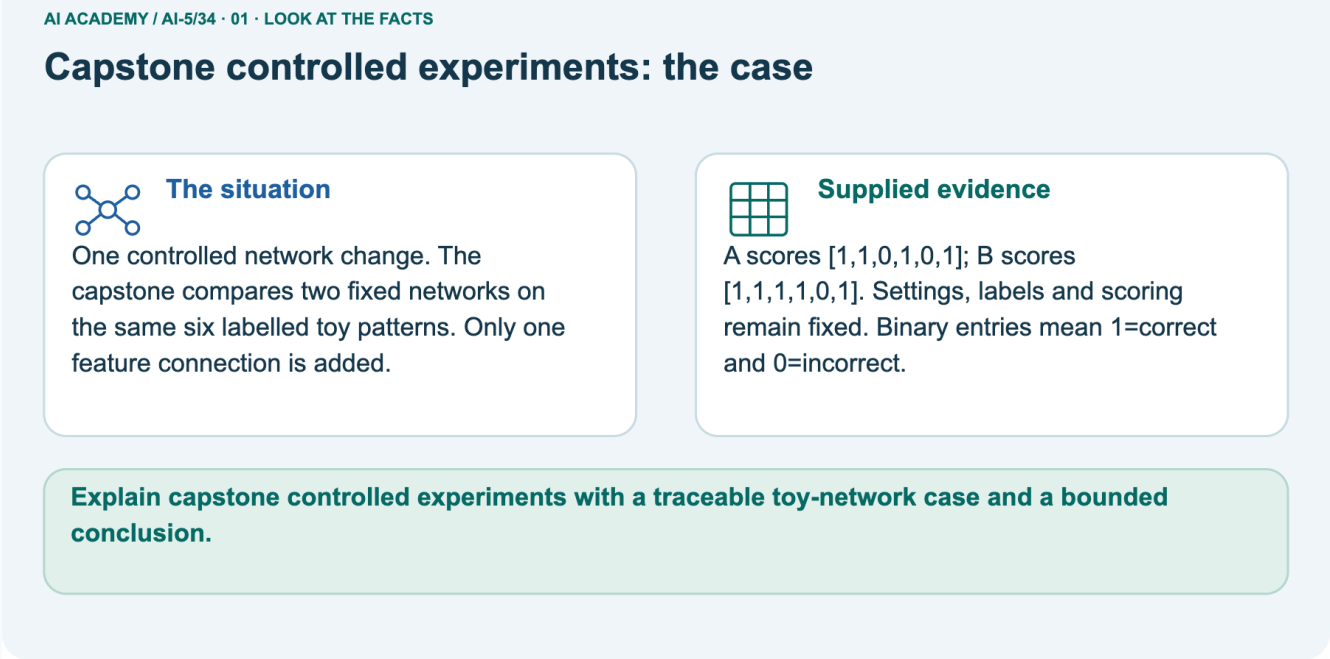
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Capstone controlled experiments picture from its labels, then record a relevant limit.

01. Capstone controlled experiments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

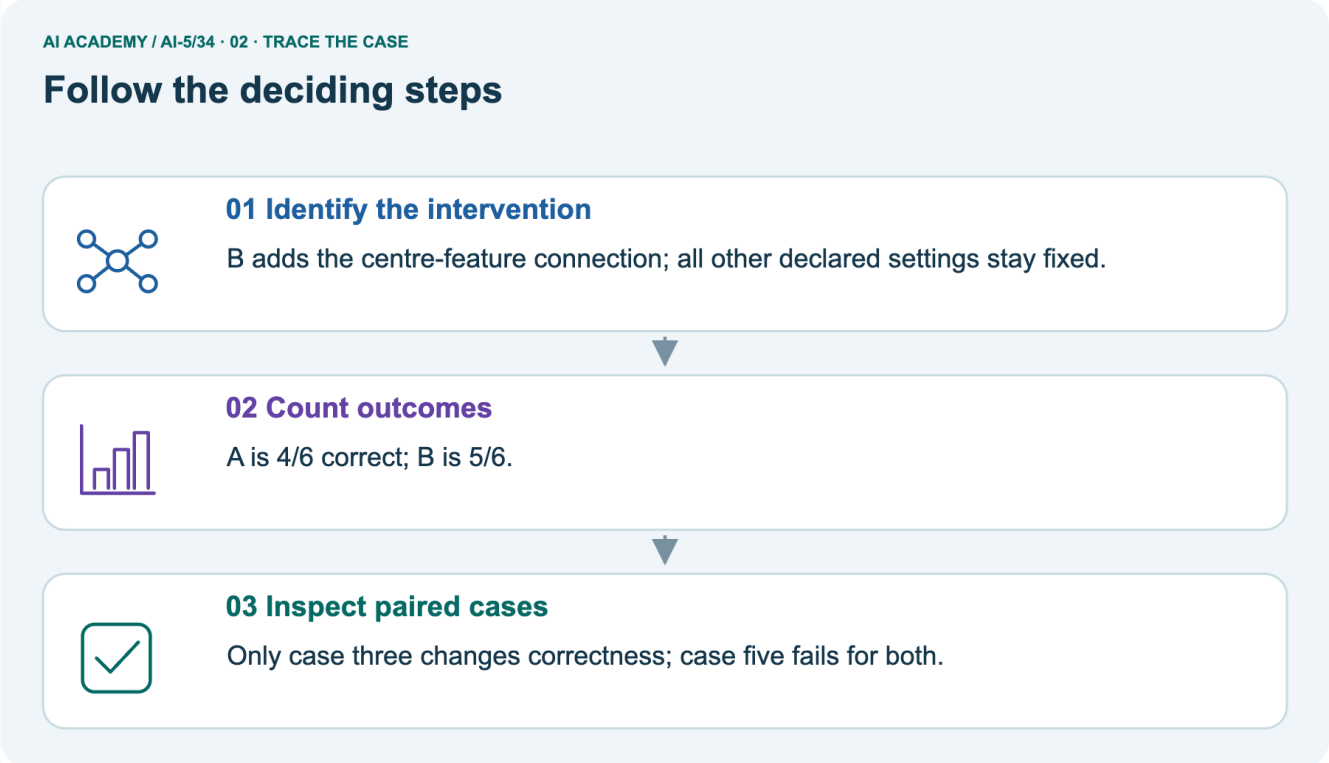
One controlled network change. The capstone compares two fixed networks on the same six labelled toy patterns. Only one feature connection is added. Supplied facts: A scores [1,1,0,1,0,1]; B scores [1,1,1,1,0,1].

Settings, labels and scoring remain fixed. Binary entries mean 1=correct and 0=incorrect.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



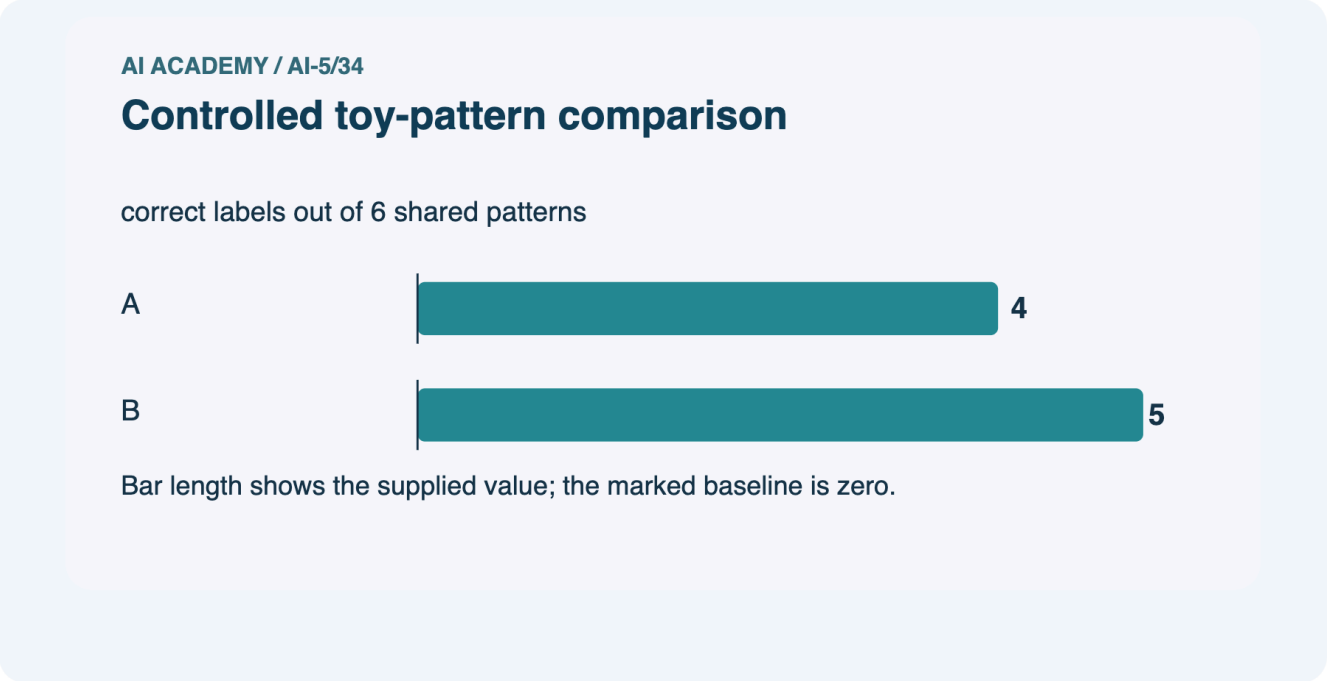
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the intervention: B adds the centre-feature connection; all other declared settings stay fixed. → Count outcomes: A is 4/6 correct; B is 5/6. → Inspect paired cases: Only case three changes correctness; case five fails for both.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Controlled toy-pattern comparison



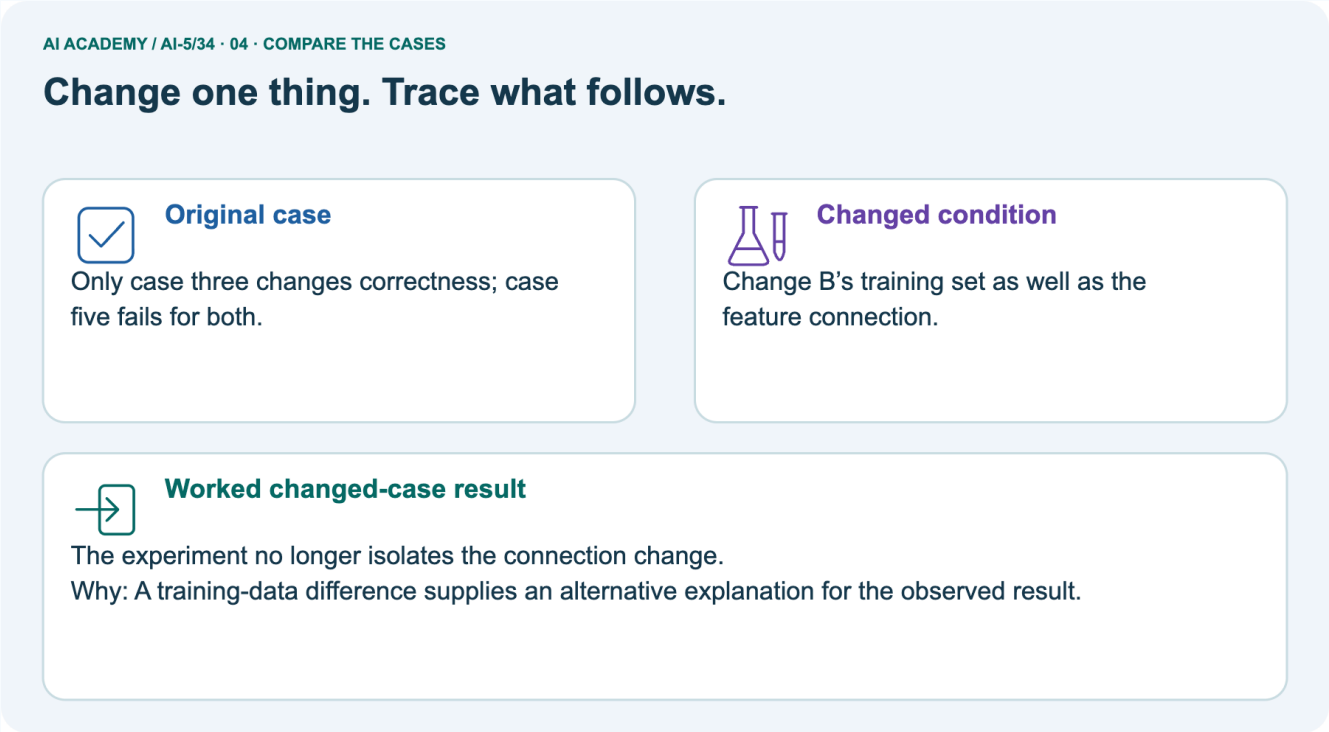
Follow the labelled picture with the matching explanation. Open the vector diagram

Controlled toy-pattern comparison. Units: correct labels out of 6 shared patterns. A: 4. B: 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

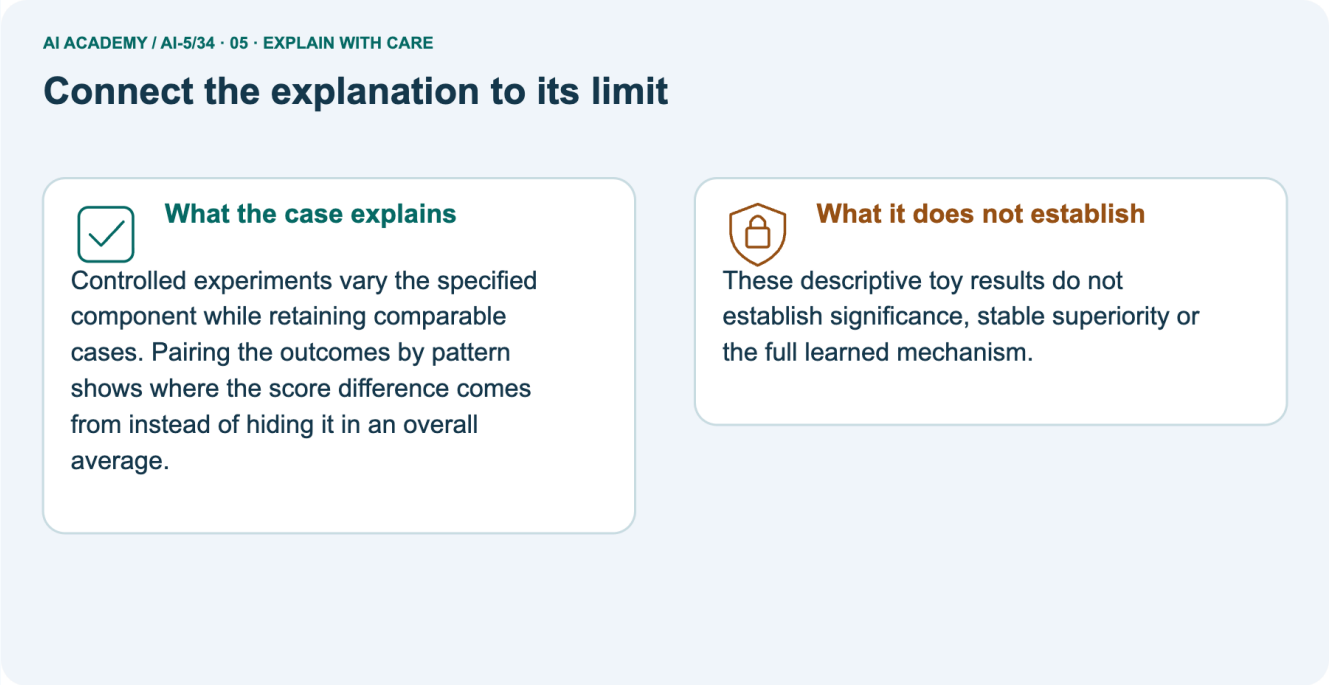
Original case: Only case three changes correctness; case five fails for both. Changed condition: Change B's **training** set as well as the feature connection.

Expected result: The experiment no longer isolates the connection change. Reason: A **training**-data difference supplies an alternative explanation for the observed result.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

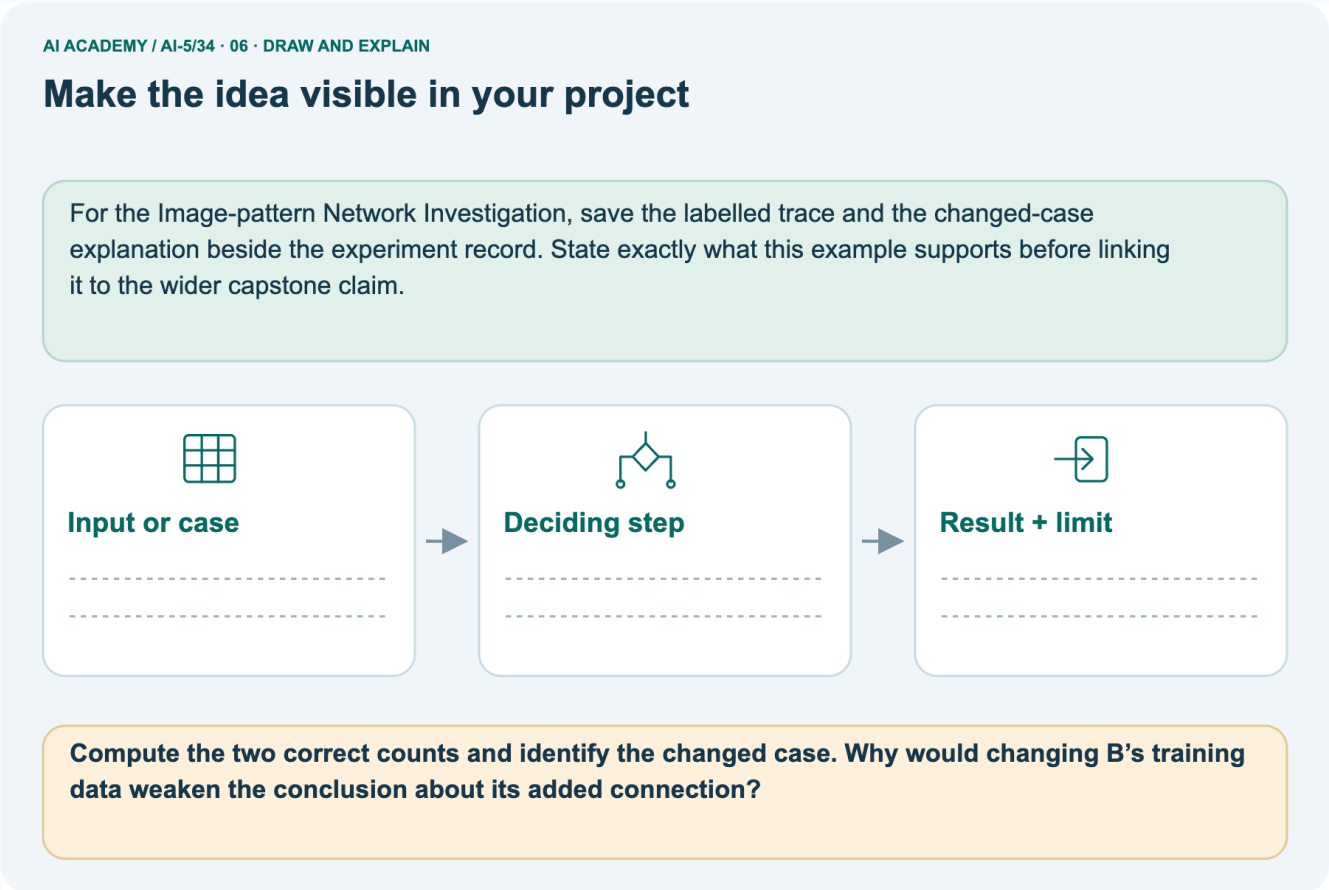
Controlled experiments vary the specified component while retaining comparable cases. Pairing the outcomes by pattern shows where the score difference comes from instead of hiding it in an overall average.

Limit: These descriptive toy results do not establish significance, stable superiority or the full learned **mechanism**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Compute the two correct counts and identify the changed case. Why would changing B's **training** data weaken the conclusion about its added connection?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.